PROJECT PHASE 1 / DATA REQUIREMENTS / TEAM 41

Wife Carrying World Championships Database

A mini-world study of the Eukonkanto competition in Sonkajärvi, Finland.

This document presents the data, integrity, and functional requirements for a database that captures the athletes, pairs, championships, race heats, results, officials, protests, teams, and sponsors of the Wife Carrying World Championships. It is grounded in the public-facing content of eukonkanto.fi and submitted as Project Phase 1.

TEAM	**Team 41 · Domain Index 6**
DOMAIN	**eukonkanto.fi · Wife-Carrying World Championship**

1. Mini World Development

This report presents the data requirements for a database that captures the operations, history, and competitive structure of the **Wife Carrying World Championships** (Finnish: *Eukonkannon maailmanmestaruuskilpailut*), an annual sporting event held in Sonkajärvi, Finland. The championship is organised by *Sonkajärven Eukonkanto Oy*, a company owned by the Sonkajärvi Entrepreneurs Association, and is governed by the rules of the International Wife Carrying Competition Rules Committee (IWCCR, also known by its Finnish abbreviation EKSVK). The official domain is eukonkanto.fi/en/front-page/.

1.1 Domain Study & Idea Formulation

The Wife Carrying World Championships were first held in 1992 as a Finnish national competition and elevated to a world championship in 1996. The event is contested annually on the first weekend of July in Sonkajärvi, Finland; the 2026 edition (3–4 July) was the 29th running, two editions having been skipped during the COVID-19 pandemic. The sport commemorates the local legend of *Herkko Ronkainen*, a 19th-century robber from Sukeva whose exploits — by legend, making off with women on his back — are re-enacted by the modern race [eukonkanto.fi/en/the-robber-ronkainen/]. The championship weekend comprises five competitive categories: the **Sprint** (Friday evening, 120 m track), the **Main Race** (Saturday afternoon, 253.5 m, open category), the **Women's Main Race**, the **40+ years** category, and the **Team Competition** (in which a team of three carriers and one carried share the 253.5 m distance). Each category is run as a sequence of heats; the best Sprint performers advance to a Sprint Final. In 2026, the Sprint Final was won by Teemu Tuovinen and Jatta Leinonen, while the Main Race was won for the ninth time by 61-year-old Taisto Miettinen (carrying Katja Kovanen)[eukonkanto.fi/en/results/,eukonkanto.fi/en/2026/07/04/taisto-miettinen-and-katja-kovanen-become-wife-carrying-world-champions/].

The 253.5 m official track is partially sandy and gravel-surfaced, with one water obstacle (~1 m deep) and two dry obstacles; the Sprint track is ~120 m with a single water obstacle. The rules [eukonkanto.fi/en/competition-track-and-rules/] require the carrier to be at least 18 years old, the carried ("wife") to be at least 17 years old, and the carried to weigh at least 49 kg — a weighted wife-carrying backpack must be worn if not. There is no penalty for dropping the wife, but the carrier must immediately pick her up. Spiked heels, hiking boots, soccer shoes, and any footwear with sharp parts are prohibited; shoes may not be taped; footwear is inspected before the race. The competition is governed by an on-site jury: a written protest must be submitted within 15 minutes of the last pair's finish, accompanied by a non-refundable €150 fee. Participation fees are €55 (individual) and €160 (team); cancellation is permitted up to 30 days before the competition for a €10 administrative fee, with no refund thereafter.

The 2026 championship drew competitors from 12 countries — and approximately 2,500 spectators [eukonkanto.fi/en/2026/06/25/wife-carrying-world-championships-to-attract-participants-from-12-different-countries/]. Press accreditation is handled via a structured form capturing outlet name, journalist role, full address, phone, email, special requests, drone usage, arrival date and time, and group member names [eukonkanto.fi/en/accreditation-press/]. The championship is the apex of an international **Official Grand Prix** circuit with sister races in China (Yulin, Guangxi), Australia (Singleton), Belgium (Dentergem), and the United States (Sunday River, Maine), each qualifying winners for the Sonkajärvi World Championship [eukonkanto.fi/en/official-grand-prix/].

1996 First World Championship	**253.5 m** Track length (Main)	**120 m** Track length (Sprint)	**12** Countries in 2026
18 yrs Min. carrier age	**17 yrs** Min. carried age	**49 kg** Min. carried weight	**€150** Protest fee

In Scope

The database captures the **competitive and operational core** of the Wife Carrying World Championships as published on eukonkanto.fi. The in-scope entities are: athletes (carriers and carried) and their nationalities; pairs (the carrier–carried unit); teams (for the team competition); championships and editions; race categories (Sprint, Main, Women's Main, 40+, Team); race heats and results; officials (jury, timekeepers, referees, announcers); protests and protest decisions; venues (Sonkajärvi Sports Field and the international Grand Prix venues); sponsors; and press accreditation. The functional side captures registration, scheduling, timing, ranking, protest handling, and historical analytics.

Out of Scope

The following are explicitly **out of scope**: (a) e-commerce features of the website (ticket sales, T-shirt sales, online store basket) — these are tracked by a separate WordPress/WooCommerce instance and are not part of the sporting database; (b) full content management of news articles (we store only a citation pointer to a news URL when one justifies a domain fact); (c) detailed accommodation, transport, parking, and area-map logistics — these are referenced only as venue attributes; (d) the Friday/Saturday evening parties (an 18+ entertainment event, not a sporting competition) — we record only that the championship has an evening programme; (e) the Robber Ronkainen dramatised legend — we treat it as background context, not as data; (f) individual spectator or attendee records (we store aggregate spectator counts per championship, not per-person data).

Why this scope

The scope boundary is drawn where the official website stops publishing structured, queryable facts. The Results page publishes ranked results per heat per category — this is the core data and is fully in scope. The Rules page publishes enforceable constraints (ages, weights, fees, protest windows) — these become integrity constraints in the database. The Programme page publishes the schedule — this becomes the heat and category schedule. The Grand Prix page publishes the international circuit — this justifies a multi-venue model. Conversely, the WooCommerce storefront, the news blog, and the marketing copy on the website are not structured sporting facts and would introduce unnecessary complexity if modelled.

Webpage Citations

The following pages of the official Wife Carrying World Championship website (eukonkanto.fi) were consulted to ground the domain study (§1.1), the integrity constraints (§2.5), and the user roles (§1.2) in what the website actually publishes. URLs are listed in the order first cited in this document.

#	URL	Used for
1	https://eukonkanto.fi/en/front-page/	General event identity; navigation structure; organiser name.
2	https://eukonkanto.fi/en/the-robber-ronkainen/	Historical origin legend; background context for §1.1.
3	https://eukonkanto.fi/en/competition-track-and-rules/	Track length (253.5 m, 120 m sprint), water/dry obstacle counts, age thresholds (18/17), weight threshold (49 kg), footwear rules, protest window (15 min), protest fee (€150), participation fee (€55/€160), cancellation policy (€10, 30 days), prize structure.
4	https://eukonkanto.fi/en/results/	Heat-by-heat results for Sprint, Main, Women's Main, 40+, Team categories; bib numbers; pair-vs-pair timing differentials; country representations (used to derive the 12-country figure).
5	https://eukonkanto.fi/en/2026/07/04/taisto-miettinen-and-katja-kovanen-become-wife-carrying-world-champions/	2026 Main Race winner (Taisto Miettinen, 9th title); 2026 Team Competition winner (Team Beer for my Horses); spectator count (~2,500); father-son podium (Joni & Jori Juntunen).
6	https://eukonkanto.fi/en/2026/06/25/wife-carrying-world-championships-to-attract-participants-from-12-different-countries/	12-country participation figure; 60 main-race couples; 6 team-competition teams; 4 women's-category couples; 10 sprint couples; live-stream partner; 30-year inaugural anniversary; COVID skip years (2).
7	https://eukonkanto.fi/en/participating-competitors-and-countries/	Confirmed competitor names by category; countries represented (Finland, Estonia, Lithuania, USA, UK, Singapore, Ireland, Philippines, Slovakia, Czech Republic, Ukraine, Russia, Germany, France, Italy, Serbia, Sweden, Austria).
8	https://eukonkanto.fi/en/guidance-for-competitors/	On-site registration windows; competitor wristband issuance; bib number distribution; practice schedule; parade time; awards time; registration closing deadline.
9	https://eukonkanto.fi/en/programme/	Two-day programme structure; Sprint Friday evening, Team Race Saturday 1 pm, Main Race Saturday 3 pm, awards 5:30 pm; market place hours.
10	https://eukonkanto.fi/en/official-grand-prix/	Grand Prix sister events: China (Yulin, Guangxi), Australia (Singleton), Belgium (Dentergem), USA (Sunday River, Maine); continental grouping (Asia, Australia, Europe, North America).
11	https://eukonkanto.fi/en/wife-carrying-world-championship-registration/	Registration deadline (Wednesday before competition, 10 pm); individual fee €55; team fee €160; payment via online store.
12	https://eukonkanto.fi/en/accreditation-press/	Press accreditation form fields: media name, person, job title, address, phone, email, special requests, drone permission, arrival date/time, group member names.
13	https://eukonkanto.fi/en/sustainability/	Founding year (1992 Finnish, 1996 World); COVID skip years; organising entity (Sonkajärven Eukonkanto Oy); ownership (Sonkajärvi Entrepreneurs Association); security rules (forbidden items list).

14	`https://eukonkanto.fi/en/contact/`	Organiser contact details: Anni Kainulainen (office), Sanna-Mari Nuutinen (international interviews), marketplace inquiries; legal address Lyseotie 1, 74300 Sonkajärvi.
15	`https://eukonkanto.fi/en/events/`	Upcoming events list structure; evening-party pricing (€38); adults-only (18+) age restriction.

1.2 User Identification & Interaction

The database is designed to serve seven distinct classes of users, each with a different interaction profile. Identifying them up front determines which functional requirements are mandatory and which are read-only.

#	User class	Primary interactions with the database
1	**Competitors** (carriers and carried)	Register themselves, form pairs, sign up for categories, view their own heat schedule and historical results, file protests against results.
2	**Team Captains**	Compose a team (3 carriers + 1 carried), register the team for the team competition, substitute members before the registration deadline, view team results.
3	**Officials** (jury, timekeepers, referees, announcers)	Record race timings and DQ statuses, submit jury decisions on protests, inspect footwear and equipment, manage heat scheduling.
4	**Event Organisers** (Sonkajärven Eukonkanto Oy staff)	Create a new championship edition, configure categories and fees, accredit press, assign officials to categories, manage sponsor contracts, view aggregate analytics.
5	**Press & Media**	Apply for accreditation, view real-time results, search historical archives, generate country- and athlete-level statistics for reporting.
6	**Spectators / Public**	View live and historical results, search athletes by name or country, view championship standings and statistical leaders.
7	**System Administrators**	Manage integrity constraints, perform schema migrations, audit protest-handling logs, archive old championships, restore from backups.

Table 1.1 — User classes and their primary interactions with the database.

The first three classes (competitors, team captains, officials) are **transactional** users: they insert and modify data through forms validated by integrity constraints. The last four are primarily **read-only** users whose access is mediated by retrieval and analysis operations. The functional requirements in Section 3 are derived from this user map.

1.3 Purpose of the Database

A database is the correct solution for this mini-world because the data is **multi-entity, multi-relationship, and historically accumulating**. A spreadsheet or a flat-file solution fails on at least four counts.

(1) Multi-entity integrity. A pair consists of two athletes; a team consists of four athletes; a heat has many pairs and a pair runs in many heats; a championship has many categories; a protest is filed by a pair against a result, decided by an official, attached to a heat. These relationships require foreign-key enforcement and cascading updates — capabilities that spreadsheets do not natively provide. A change of citizenship for one athlete, for example, must propagate to every pair, team, and result that athlete has ever been part of. Without enforced referential integrity, the historical record would silently drift.

(2) Constraint enforcement at the data layer. The official rules define precise constraints: carrier age ≥ 18, carried age ≥ 17, carried weight ≥ 49 kg (or backpack), bib number unique within a championship, protest submission within 15 minutes of last finish, participation fee €55 / €160, non-refundable €150 protest fee, €10 cancellation fee with 30-day refund window. In a database these are CHECK constraints, triggers, and stored procedures; in a flat file they are at best manual checks, at worst silently violated.

(3) Historical analytics across 30+ years. The championship has been run since 1996. Generating the all-time medal table, the country with the most golds, the athlete with the most appearances, or the average finishing-time trend requires joining pairs, results, championships, and athletes across decades. A relational database makes these queries first-class; a spreadsheet solution would require perennial re-merging of files and would scale poorly past a few hundred rows.

(4) Multi-user concurrent access. During the championship weekend, officials record timings in real time, journalists query results, organisers monitor registrations, and competitors check their heat schedule — all simultaneously. A database provides transactional isolation and concurrent read access; a shared spreadsheet or file-based solution would either lock up or corrupt under concurrent writes.

In summary, the database offers what non-DB solutions cannot: declarative integrity constraints, multi-entity joins, multi-year analytics, and multi-user concurrency. These four capabilities are not incidental — they are **load-bearing** for the championship's operational correctness and historical record.

2. Database Requirements

This section specifies the entity types, relationship types, attribute types, and integrity constraints that constitute the conceptual schema of the Wife Carrying World Championships database.

2.1 Entity Types

The schema contains **nine regular entity types** (PERSON, COUNTRY, VENUE, TEAM, CHAMPIONSHIP, RACE_CATEGORY, SPONSOR, PRESS_OUTLET, ORGANISER) and **three weak entity types** (RACE_HEAT, RACE_RESULT, PROTEST). Two of the regular entity types (COMPETITOR, OFFICIAL) are subclasses of PERSON. Each entity is described below with its key attributes; full attribute catalogues appear in Appendix A.

2.1.1 PERSON

PERSON is the master entity for every human referenced by the database: athletes, officials, organisers, and press contacts. It carries a primary key, an alternate key (email), and a super key — together demonstrating the full key hierarchy used in this schema.

Attribute	Type	Key role	Notes
person_id	INT	**Primary key**	System-generated surrogate; auto-increment.
email	VARCHAR(255)	**Alternate key** (unique, NOT NULL)	Unique email. Serves as login identifier for self-service.
full_name	COMPOSITE	n/a	Composite attribute: {first_name, last_name}. See §2.4.

date_of_birth	DATE	n/a	Used to derive age; age must satisfy role-specific minimums.
phone_numbers	MULTI-VALUED	n/a	Each person may have zero or more phones (mobile, landline, emergency).
country_code	CHAR(2)	Foreign key → COUNTRY	Nationality. ISO 3166-1 alpha-2.
gender	ENUM	n/a	Values: M, F, X, unspecified.
age (derived)	INT	Derived	Computed as floor((today − date_of_birth) / 365.25).
{person_id, date_of_birth}	COMPOSITE KEY	**Super key**	Trivially unique combination — demonstrates the super key concept.

Table 2.1 — PERSON entity, carrying a primary key, an alternate key, and a super key.

2.1.2 COMPETITOR — subclass of PERSON (IS-A, disjoint)

COMPETITOR is a subclass of PERSON. The IS-A relationship is **disjoint** with OFFICIAL (a person is registered either as a competitor or as an official for a given championship, never both). The discriminating attribute is *role_type* ∈ {COMPETITOR, OFFICIAL}. A COMPETITOR inherits all PERSON attributes and adds competitor-specific attributes: *bib_number* (unique within a championship), *role* ∈ {CARRIER, CARRIED} (some competitors can serve in either role across different pairs), *weight_kg* (recorded at weigh-in for the carried role, since the 49 kg minimum applies), and *career_start_year*.

2.1.3 OFFICIAL — subclass of PERSON (IS-A, disjoint)

OFFICIAL is the second subclass of PERSON, disjoint with COMPETITOR. An OFFICIAL has *official_role* ∈ {JURY, REFEREE, TIMEKEEPER, ANNOUNCER, EQUIPMENT_INSPECTOR, SECURITY_CHIEF} and *certification_level* ∈ {1, 2, 3}. The jury role is the only one authorised to record PROTEST_DECISION records (see §2.3).

2.1.4 COUNTRY

COUNTRY represents a nationality. Its primary key is the ISO 3166-1 alpha-2 code (e.g., FI, US, EE, LT, UK, SG, IE, PH, SK, CZ, UA, DE, FR, IT, RS, SE, AT). Other attributes: *name*, *continent* ∈ {ASIA, EUROPE, NORTH_AMERICA, SOUTH_AMERICA, AFRICA, OCEANIA}, and *first_participation_year* (derived — the year of the earliest championship in which any competitor from this country participated).

2.1.5 VENUE

A VENUE is a physical location where a championship or Grand Prix event is held. The primary key is *venue_id* (surrogate). Attributes include *name*, *address* (composite: {street, city, postal_code, country_code}), *capacity*, and *track_length_m* (the 253.5 m figure for Sonkajärvi; 120 m for sprint-only venues).

2.1.6 TEAM

A TEAM is a unit of 3 carriers + 1 carried that competes in the Team Competition. The primary key is *team_id*. Attributes: *name* (e.g., "Team Beer for my Horses", "Team Big Beef", "Team 2 SPOLK"), *founded_year*, *home_country_code*. The recursive relationship TEAM_PARTNERSHIP (see §2.2.9) allows teams to be affiliated with a parent national team.

2.1.7 CHAMPIONSHIP

CHAMPIONSHIP represents a single annual edition of the Wife Carrying World Championship (or, in extension, a Grand Prix event). The primary key is *champ_id*. Notable attributes: *year* (e.g., 2026), *edition_number* (29 in 2026), *start_date*, *end_date*, *host_venue_id* (FK → VENUE), *organiser_id* (FK → ORGANISER), *spectator_count*, *is_grand_prix* (Boolean — distinguishes the World Championship from a Grand Prix qualifier), and *years_since_inaugural* (derived — year − 1996).

2.1.8 RACE_CATEGORY

RACE_CATEGORY is a catalogue entity defining the categories contested in any championship: SPRINT (120 m), MAIN (253.5 m open), WOMENS_MAIN (253.5 m, female-female pairs), MASTERS_40 (40+ years), TEAM (3 carriers + 1 carried). Primary key *cat_code*. Attributes: *name*, *distance_m*, *min_age_carrier*, *min_age_carried*, *min_weight_carried_kg*, *has_water_obstacle*, *has_dry_obstacles*, *max_team_size* (NULL for non-team categories; 4 for TEAM).

2.1.9 SPONSOR and PRESS_OUTLET

SPONSOR captures commercial sponsors of the championship (e.g., DNA, the 2026 telecommunications partner). Primary key *sponsor_id*. PRESS_OUTLET captures media organisations applying for accreditation. Primary key *outlet_id*. Both have a composite *contact* attribute {email, phone, address}.

Entity & Attribute Catalogue

Entity	Type	Key attributes	Other notable attributes
PERSON	Regular	person_id (PK); email (alternate key); {person_id, date_of_birth} (super key)	full_name (composite), date_of_birth, phone_numbers (multi-valued), country_code (FK), gender, age (derived)
COMPETITOR	Subclass of PERSON (disjoint)	inherits person_id	bib_number, role ∈ {CARRIER, CARRIED}, weight_kg, career_start_year
OFFICIAL	Subclass of PERSON (disjoint)	inherits person_id	official_role, certification_level
COUNTRY	Regular	country_code (PK, ISO alpha-2)	name, continent, first_participation_year (derived)
VENUE	Regular	venue_id (PK)	name, address (composite), capacity, track_length_m
TEAM	Regular	team_id (PK)	name, founded_year, home_country_code (FK)
CHAMPIONSHIP	Regular	champ_id (PK)	year, edition_number, start_date, end_date, host_venue_id (FK), organiser_id (FK), spectator_count, is_grand_prix, host_languages (multi-valued), years_since_inaugural (derived)
RACE_CATEGORY	Regular	cat_code (PK)	name, distance_m, min_age_carrier, min_age_carried, min_weight_carried_kg, has_water_obstacle, has_dry_obstacles, max_team_size

SPONSOR	Regular	sponsor_id (PK)	name, country, sponsorship_categories (multi-valued), contact (composite)
PRESS_OUTLET	Regular	outlet_id (PK)	name, country, contact (composite)
ORGANISER	Regular	organiser_id (PK)	name, legal_form, founded_year
RACE_HEAT	Weak (owners: CHAMPIONSHIP × RACE_CATEGORY)	(champ_id, cat_code, heat_number) — composite	scheduled_time, status ∈ {SCHEDULED, IN_PROGRESS, COMPLETED, CANCELLED}
RACE_RESULT	Weak (owners: RACE_HEAT × PAIR)	(champ_id, cat_code, heat_number, carrier_id, carried_id) — composite	finish_time_ms, diff_to_leader_ms, drop_count, finish_status ∈ {COMPLETED, DNF, DQ, UNDER_PROTEST}, rank_in_heat, rank_in_category (derived)
PROTEST	Weak (owner: CHAMPIONSHIP)	(champ_id, protest_seq) — composite	filed_at, filed_by_pair_id, against_pair_id, against_heat_id, fee_paid, description, status ∈ {PENDING, UPHELD, REJECTED, WITHDRAWN}
PROTEST_DECISION	Weak (owner: PROTEST)	(champ_id, protest_seq, decision_seq) — composite	decision_text, decided_at, decided_by_official_id (FK), decision_outcome
PAIR	Associative entity (from R1 CARRIES)	(carrier_id, carried_id) — composite	formed_year, is_active, weight_deficit_kg (derived)

2.2 Weak Entity Types

The schema contains **three weak entity types**. Each is existence-dependent on one or more identifying owner entity types and uses a composite key consisting of the owner's primary key plus a partial key.

Weak entity	Identifying owner(s)	Partial key	Full (composite) key
RACE_HEAT	CHAMPIONSHIP, RACE_CATEGORY	heat_number (1, 2, 3, …)	(champ_id, cat_code, heat_number)
RACE_RESULT	RACE_HEAT, PAIR (the CARRIES relationship instance)	finish_time_ms	(champ_id, cat_code, heat_number, carrier_id, carried_id)
PROTEST	CHAMPIONSHIP	protest_seq (1, 2, 3, … within championship)	(champ_id, protest_seq)

Table 2.2 — The three weak entity types and their identifying owners.

A **RACE_HEAT** cannot exist without a championship and a category — there is no "orphan" heat floating without a championship. Its identity is the triple (championship, category, heat number within category). A **RACE_RESULT** cannot exist without both a heat and a pair: it records the time and rank of one specific pair in one specific heat. A **PROTEST** cannot exist outside a championship; it is identified by (championship, sequence number).

2.3 Relationship Types

The schema defines **ten relationship types**. Each is listed below with its degree, cardinality ratio, and participation constraints. Cardinality is given in the min..max notation on each side.

#	Relationship	Participating entities	Cardinality	Participation
R1	**CARRIES** *(recursive — same entity in distinct roles)*	COMPETITOR (role: carrier) ↔ COMPETITOR (role: carried)	M:N	Partial on both sides — a competitor may compete in many pairs across years; a pair has exactly one carrier and one carried.
R2	**PARTICIPATES_IN**	PAIR (associative entity of R1) ↔ RACE_HEAT	M:N	A pair may run in many heats (across championships); a heat has 2–many pairs (a heat is cancelled if fewer than 2 pairs register).
R3	**HOSTS**	VENUE (1) ↔ CHAMPIONSHIP (N)	1:N	Total on CHAMPIONSHIP side (every championship has exactly one host venue); partial on VENUE side (some venues have never hosted).
R4	**HAS_CATEGORY**	CHAMPIONSHIP ↔ RACE_CATEGORY	M:N	A championship typically has 4–5 categories; a category (e.g., MAIN) appears in many championships. Partial on both sides.
R5	**COMPETES_FOR**	COMPETITOR ↔ TEAM	M:N	A competitor may belong to zero or more teams over their career; a team has exactly 3 carriers + 1 carried at any single championship.
R6	**SPONSORS** *(with descriptive attrs: tier, contract_start, contract_end)*	SPONSOR ↔ CHAMPIONSHIP	M:N	A sponsor may sponsor many championships; a championship has 1–many sponsors. The relationship instance carries tier ∈ {TITLE, GOLD, SILVER, BRONZE}.
R7	**JUDGES** *(ternary, n = 3)*	OFFICIAL × CHAMPIONSHIP × RACE_CATEGORY	M:N:N	An official may judge many (championship, category) pairs; each (championship, category) is judged by ≥ 1 official. Total on (CHAMPIONSHIP × RACE_CATEGORY) side.
R8	**ACCREDITS** *(with descriptive attrs: accreditation_year, drone_permission)*	PRESS_OUTLET ↔ CHAMPIONSHIP	M:N	An outlet may cover many championships; a championship may accredit many outlets. Partial on both.
R9	**REPRESENTS**	PAIR ↔ COUNTRY	M:N	A pair may represent one or two countries (e.g., Team Beer for my Horses: USA/LTU); a country has many pairs.
R10	**OWNS**	ORGANISER (1) ↔ CHAMPIONSHIP (N)	1:N	Total on CHAMPIONSHIP — every championship is owned by exactly one organiser (Sonkajärven Eukonkanto Oy for the World Championships; the local Grand Prix organiser for GP events).

Table 2.3 — The ten relationship types, including the recursive CARRIES and the ternary JUDGES.

2.3.1 Alternative Modelling of the Ternary JUDGES

The ternary JUDGES(OFFICIAL, CHAMPIONSHIP, RACE_CATEGORY) can be re-expressed as **three binary relationships plus a weak entity**:

- **JUDGES_OC(OFFICIAL, CHAMPIONSHIP)** — records that an official is accredited for a championship (M:N).
- **JUDGES_OR(OFFICIAL, RACE_CATEGORY)** — records that an official is qualified to judge a category (M:N).
- **CHAMP_CATEGORY(CHAMPIONSHIP, RACE_CATEGORY)** — weak entity owned half by CHAMPIONSHIP and half by RACE_CATEGORY; its existence captures the fact that "this championship actually contests this category this year" (since not every championship contests every category).
- **JUDGES_OC_CAT(OFFICIAL, CHAMP_CATEGORY)** — a binary relationship between OFFICIAL and the CHAMP_CATEGORY weak entity, recording "this official judges this championship's instance of this category".

The ternary form is preferred because it is more compact and matches the real-world semantics (an official is assigned to a championship-category pair, not to a championship or a category in isolation). The binary decomposition is useful for performance or for distributed storage where CHAMP_CATEGORY is materialised as a bridging table. The decomposition does lose information: the ternary form unambiguously says "official O judges category C in championship Y"; the binary form allows the inconsistent state where O is qualified for C and accredited for Y but never explicitly assigned to (Y, C).

2.4 Composite, Multi-Valued & Derived Attributes

The schema contains **three composite attributes, three multi-valued attributes, and three derived attributes**.

Type	Attribute	Owner	Components / derivation
Composite	full_name	PERSON	{first_name, last_name}
Composite	address	VENUE	{street, city, postal_code, country_code}
Composite	contact	PRESS_OUTLET / SPONSOR	{email, phone, address}
Multi-valued	phone_numbers	PERSON	Set of phone strings; e.g., {+358 40 1234567, +358 50 9876543}
Multi-valued	host_languages	CHAMPIONSHIP	Set of languages used at the event; e.g., {Finnish, English} (Sonkajärvi 2026) or {English, Chinese} (Yulin Grand Prix)
Multi-valued	sponsorship_categories	SPONSOR	Set of categories the sponsor supports; e.g., {MAIN, SPRINT, TEAM}
Derived	age	PERSON	floor((today − date_of_birth) / 365.25)
Derived	weight_deficit_kg	PAIR (or COMPETITOR in carried role)	max(0, 49 − carried_weight_kg). Determines backpack weight needed.

Derived	rank_in_category	RACE_RESULT	Rank of finish_time within all results of the same (championship, category). Recomputed on every insert / update.
Derived	years_since_inaugural	CHAMPIONSHIP	year − 1996 (the inaugural World Championship year)
Derived	first_participation_year	COUNTRY	min(year) over all championships in which any competitor from this country participated

Table 2.4 — Composite, multi-valued, and derived attributes.

2.5 Integrity Constraints & Enforcement(Side note)

Integrity constraints are the rules the database must enforce to remain a faithful model of the mini-world. They are drawn directly from the official rules at eukonkanto.fi/en/competition-track-and-rules/ and from the registration and accreditation pages.

#	Constraint	Type	Enforcement mechanism
C1	Carrier age ≥ 18 years on championship start_date	Domain / tuple CHECK	CHECK (DATEDIFF(start_date, person.date_of_birth) ≥ 18×365.25) on insertion into COMPETES_IN with role=CARRIER.
C2	Carried age ≥ 17 years on championship start_date	Domain / tuple CHECK	Same as C1 with role=CARRIED and threshold 17.
C3	Carried weight ≥ 49 kg OR backpack_weight_kg = (49 − weight_kg) > 0	Tuple CHECK + trigger	CHECK on PAIR; trigger verifies backpack is worn when weight < 49 kg.
C4	Bib number unique within (championship, category)	Unique index	UNIQUE(champ_id, cat_code, bib_number) on COMPETITOR_REGISTRATION.
C5	Protest submission within 15 minutes of last pair's finish	Trigger (before-insert)	Trigger compares NOW() with (max(finish_time) + 15 min) for the relevant (championship, heat); rejects if later.
C6	Protest fee €150 non-refundable; participation fee €55 (individual) / €160 (team)	Domain CHECK + transactional insert	CHECK on PROTEST.fee_paid = 150; CHECK on REGISTRATION.fee_paid ∈ {55, 160}.
C7	Cancellation only ≥ 30 days before championship start; €10 admin fee; no refund thereafter	Trigger (before-update on REGISTRATION.status)	Trigger computes DATEDIFF(start_date, NOW()); if < 30, blocks status change to CANCELLED unless fee forfeiture is acknowledged.
C8	No footwear with sharp parts; no taped shoes; no spiked heels / hiking boots / soccer shoes	Application-layer rule + audit log	Equipment-inspector sign-off stored as inspection_record (FK → OFFICIAL with role EQUIPMENT_INSPECTOR); CHECK that inspection_record_id IS NOT NULL before heat start.
C9	A pair must consist of two distinct PERSONs	Tuple CHECK	CHECK (carrier_id ≠ carried_id) on PAIR.
C10	A team has exactly 3 carriers + 1 carried (4 distinct members)	Trigger + COUNT assertion	After-insert trigger on TEAM_MEMBER verifies role counts.

C11	An official judging a (championship, category) must have certification_level ≥ 2; jury role requires certification_level = 3	Tuple CHECK + cross-table	CHECK on JUDGES relationship instance joined with OFFICIAL.
C12	Referential integrity: every FK must reference an existing PK	Declarative FK	FOREIGN KEY ... REFERENCES ... on all relationships. ON DELETE RESTRICT for championship-related FKs (no championship deletion while references exist); ON DELETE SET NULL for optional relationships.
C13	A protest is filed by a pair registered in the same championship; decision recorded only by an OFFICIAL with role = JURY	Cross-table CHECK + trigger	CHECK (protest.pair.championship = protest.championship); trigger on PROTEST_DECISION insert verifies decider.official_role = JURY.
C14	No data redundancy: pair (carrier, carried) is stored once; results reference the pair, not the two athletes separately	Schema design (normalisation)	RACE_RESULT references PAIR via FK; never stores carrier_id and carried_id directly. This is the standard normalisation rule against the same fact being stored twice.
C15	Cancellation fee (€10) charged only if registration is cancelled within the refund window; participation fee forfeited outside the window	Tuple CHECK + computed column	CHECK on REGISTRATION.refund_amount: refund = fee_paid − 10 if cancelled ≥ 30 days before; refund = 0 otherwise.

Table 2.5 — Integrity constraints and their enforcement mechanisms. Each is derived from a specific rule on eukonkanto.fi.

These constraints collectively enforce every operational rule published on the official website. They are not abstract — each maps to a sentence in the rules page, the registration page, or the press accreditation page. For example, C5 (15-minute protest window) is a direct quotation of the rule "a written protest must be submitted to the jury within 15 minutes of the arrival of the last pair at the finish line".

3. Functional Requirements

A change to the data requirements is introduced in §3.3: the original schema tracks only finish_time_ms in RACE_RESULT. To support the rank recomputation cascade required by the update operations, we add a derived column *rank_in_category* to RACE_RESULT.

3.1 Retrieval Operations

3.1.1 Selection

> *"Retrieve the complete data rows of all race results recorded in the 2026 Wife Carrying World Championship Main Race category."*

This query selects *all* attributes of RACE_RESULT rows where the parent RACE_HEAT belongs to the (championship year = 2026, category = MAIN) pair. It returns every column — bib number, carrier name, carried name, country, finish time, rank, diff to leader, drop count, DQ status — for every pair in every heat of the Main Race. This is the raw data the Results page on eukonkanto.fi displays in its "JOUKKUE" and "W" tables.

3.1.2 Projection

> *"Display the names and countries of all carriers who participated in the Sprint category in 2026."*

This query projects only two columns — competitor full name and country name — from the join of PERSON, COMPETITOR, PAIR, PARTICIPATES_IN, RACE_HEAT, and RACE_CATEGORY, restricted to cat_code = SPRINT and champ.year = 2026 and competitor.role = CARRIER. Duplicates are removed (a carrier who ran in both a heat and the final appears once).

3.1.3 Aggregate

> *"For the 2026 Main Race, compute the maximum, minimum, and average finish time across all heats, and the count of pairs that finished without dropping the wife."*

This query applies MAX, MIN, AVG, and COUNT aggregate functions to RACE_RESULT.finish_time_ms, grouped by championship and category. The COUNT is restricted to results with drop_count = 0. The result is a single row of four scalars — typical of a stat-block on the Results page.

3.1.4 Search (partial text match)

> *"Search for athletes whose last name contains 'Miettinen'."*

This query performs a partial-text match (LIKE '%Miettinen%') against PERSON.full_name.last_name. It would return Taisto Miettinen and any other competitor whose surname contains the substring. The query is the foundation of the search box on the Participating Competitors page.

3.1.5 Search (partial text match — team names)

> *"Search for teams whose name contains 'Beer'."*

This query performs a partial-text match against TEAM.name. It would return "Team Beer for my Horses" (the 2026 team-competition winner). The query also demonstrates substring search across an entity distinct from PERSON — the same pattern generalises to searching press outlet names, sponsor names, and venue names.

3.2 Analysis Reports

Four analysis reports are specified. Each joins at least three entities and uses aggregate or window functions — they are not simple selections from a single entity.

3.2.1 Report: country medal table across all main races

> *"For each country, count the number of gold, silver, and bronze medals won in the Main Race across all championships since 1996. Rank countries by total golds, then total silvers, then total bronzes."*

This report joins COUNTRY ↔ PERSON ↔ COMPETITOR ↔ PAIR ↔ RACE_RESULT ↔ RACE_HEAT ↔ CHAMPIONSHIP ↔ RACE_CATEGORY, filters to cat_code = MAIN and rank_in_category ∈ {1, 2, 3}, then groups by COUNTRY and pivot-counts by rank. The result is the all-time medal table — a classic off-season press story.

3.2.2 Report: pairs who competed in both Sprint and Main Race in the same championship

> *"For each championship since 2015, list the pairs that competed in both the Sprint and the Main Race, with their rank in each category and the rank differential. Order by championship year descending, then by main-race rank ascending."*

This report self-joins RACE_RESULT twice on PAIR — once filtered to cat_code = SPRINT, once to cat_code = MAIN — within each championship. It computes the differential main_rank − sprint_rank to identify pairs whose endurance favours either the short or long distance. A negative differential indicates a stronger Main Race performance.

3.2.3 Report: officials with the most championship assignments

> *"For each official who has judged at least 3 championships, list the championships and categories they judged, plus the count of protests they decided. Rank by total assignments descending."*

This report joins OFFICIAL ↔ JUDGES (ternary) ↔ CHAMPIONSHIP ↔ RACE_CATEGORY on one side, and OFFICIAL ↔ PROTEST_DECISION ↔ PROTEST on the other. It demonstrates a join across both sides of the ternary JUDGES relationship and an aggregation over a weak entity (PROTEST_DECISION).

3.2.4 Report: average finish-time trend per championship

> *"For each championship since 1996, compute the average Main Race finish time and the standard deviation. Identify the championship with the fastest average and the one with the tightest field (lowest standard deviation)."*

This report joins CHAMPIONSHIP ↔ RACE_HEAT ↔ RACE_RESULT, filtered to cat_code = MAIN and finish_status = COMPLETED, then aggregates AVG and STDDEV per championship. The result is a 29-row time-series table — a typical season-recap graphic for press kits.

3.3 Modification Operations

Modification operations are grouped into insertions (3.3.a), updates (3.3.b), and deletions (3.3.c). Each operation lists the integrity constraints it touches and the cascading actions it triggers.

3.3.a Insertions

Insert I-1: Register a new competitor in the 2027 championship.

A new competitor (e.g., a first-time entrant from India) is registered as a CARRIER in the MAIN category of the 2027 championship. The insert operation creates a PERSON row, a COMPETITOR row (subclass), and a COMPETITOR_REGISTRATION row linking the competitor to (championship = 2027, category = MAIN, bib_number = assigned).

Integrity constraints checked:

- C1 — carrier age ≥ 18 on 2027-07-02 (computed from PERSON.date_of_birth).
- C4 — assigned bib_number must not already exist in (2027, MAIN).
- C12 — country_code (e.g., IN) must exist in COUNTRY.
- C6 — fee_paid = 55 (individual registration) at insert time.

Cascading actions: None on insert. A confirmation email is sent (out-of-database side effect).

Insert I-2: File a protest against a race result.

A pair that finished 4th in Heat 3 of the 2026 Main Race files a protest alleging that the 3rd-place pair dropped the wife at the water obstacle without recovering. The insert creates a PROTEST row owned by championship = 2026, with protest_seq = next sequence number, filed_by_pair_id = the protesting pair, against_pair_id = the 3rd-place pair, against_heat_id = the relevant heat, fee_paid = 150.

Integrity constraints checked:

- C5 — protest submitted within 15 minutes of max(finish_time) for the heat. A before-insert trigger computes NOW() − (max_finish + 15 min); if positive, insert is rejected with error PROTEST_WINDOW_EXPIRED.
- C6 — fee_paid must equal €150 (non-refundable).
- C13 — filed_by_pair must be registered in the same championship as the protest; against_pair must be in the same heat.
- C12 — both pair_id values must exist in PAIR.

Cascading actions: A PROTEST_DECISION slot is reserved (status = PENDING). The protested result is flagged as "under protest" (a derived display attribute) until the jury records a decision.

3.3.b Updates

Update U-1: Correct a race result finish_time after a timing-gate malfunction.

The timing gate at the water obstacle malfunctioned during Heat 2 of the 2026 Sprint. Manual review determines the actual finish time of pair #78 (Teemu Tuovinen / Jatta Leinonen) is 35.80 s, not 36.10 s as recorded. The update modifies RACE_RESULT.finish_time_ms.

Cascading actions enforced by triggers:

- **rank_in_heat** is recomputed for all results in the same heat — pair #78 moves from 2nd to 1st, the previous 1st-place pair moves to 2nd, and diff_to_leader is recomputed for every row.
- **rank_in_category** (derived attribute) is recomputed across all heats of (championship, SPRINT) — affects who qualifies for the Sprint Final.
- If the corrected result changes the Sprint Final qualifiers, a notification is queued to the displaced pair (out-of-database side effect).
- If the affected pair has a pending protest (C13), the protest's status is automatically re-evaluated by the trigger.

Integrity constraints checked: C12 (FK still valid), C5 (re-evaluate protest windows for any pending protests).

Update U-2: Change a competitor's role from CARRIER to CARRIED.

A competitor who registered as a carrier sustains an injury before the championship. Another registered competitor agrees to take the carrier role, and the injured competitor switches to the carried role. The update modifies COMPETITOR_REGISTRATION.role.

Cascading actions enforced by triggers:

- All PAIR rows where the competitor was the carrier are invalidated (status = REPLACED); a new PAIR row is created with the new carrier.

- All RACE_RESULT rows of the invalidated pairs are flagged as "team composition changed post-registration" — they remain visible in archives but are excluded from medal aggregation queries.
- C3 (carried weight ≥ 49 kg) is re-checked against the new carried competitor's weight_kg; if violated, a backpack record is auto-created.
- C10 (team composition 3 carriers + 1 carried) is re-verified if the competitor was a team member.

Integrity constraints checked: C2 (carried age ≥ 17), C3 (carried weight), C9 (pair has two distinct members), C10 (team role counts).

Update U-3: Jury upholds a protest — revert the protested result.

The jury reviews PROTEST #2026-03 (filed in I-2) and decides to uphold it: the 3rd-place pair is disqualified. The update modifies PROTEST.status = UPHELD and inserts a PROTEST_DECISION row (decision_text = "DQ for unrecovered drop", decided_by = JURY_OFFICIAL_ID, decided_at = NOW()).

Cascading actions enforced by triggers:

- RACE_RESULT.finish_status for the protested pair is set to DISQUALIFIED.
- **rank_in_heat** is recomputed — the 4th-place pair (the protester) is promoted to 3rd, the 5th-place pair is promoted to 4th, and so on.
- **rank_in_category** (derived attribute) is recomputed — affects the Sprint Final qualifiers (if a Sprint protest) or the Main Race medal assignments.
- If the protested pair was on the podium, the medal-assignment table is updated and a notification is queued to the newly-promoted pair.
- The €150 protest fee remains non-refundable (C6); no financial cascade.

Integrity constraints checked: C13 (decider must be JURY role), C5 (decision must be within 60 minutes of protest filing — a soft SLA constraint), C12.

3.3.c Deletion

Delete D-1: Delete a championship that was cancelled (e.g., 2020 COVID-19 cancellation).

The 2020 championship was cancelled due to the COVID-19 pandemic. The organisers wish to remove the empty placeholder CHAMPIONSHIP row to keep historical analytics clean. The delete attempts to remove the row from CHAMPIONSHIP where year = 2020.

Referential-integrity behaviour (C12 enforcement):

- If any RACE_HEAT exists for the championship, deletion is REJECTED (ON DELETE RESTRICT) — a championship with heats cannot be deleted; it must be archived instead.
- If any PROTEST exists for the championship, deletion is REJECTED — protests are part of the historical record.
- If any PRESS_ACCREDITATION or SPONSORSHIP exists, deletion is REJECTED unless those records are first moved to an archive table.
- If the championship has only the placeholder row (no heats, no protests, no accreditations, no sponsorships), deletion is permitted; the row is removed and any nullable FK references (e.g., CHAMPIONSHIP.first_hosted_venue_id) in other tables are SET NULL.

- The deletion is logged in an audit table (championship_id, year, deleted_by, deleted_at, reason) — this is a soft-delete supplement to the hard delete, allowing recovery if a mistake is later discovered.

This deletion illustrates the referential-integrity principle: a championship is the root of a deep dependency tree (heats → results → protests → decisions; sponsorships; accreditations; official assignments). Deleting it requires either (a) cascading down through the tree, or (b) rejecting the delete and archiving instead. The schema chooses (b) — championships are never hard-deleted while sub-records exist; the archive-then-delete pattern is the recommended operational procedure.